

Deepak Mallampati

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US-based (willing to relocate to SF/NYC), F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer who ships production agentic systems in complex, constrained environments. Strong TypeScript and Python proficiency with a track record of building AI pipelines that work under real-world constraints: ~30% fewer post-deployment defects, ~25% improved agent stability, >30% hallucination reduction. Comfortable debugging distributed systems, integrating third-party APIs with sparse documentation, and owning deliverables end-to-end. Ready to bring that mindset to Pallet's logistics AI deployment challenges.

CORE COMPETENCIES

Agentic AI Deployment • TypeScript & Node.js • Python • REST API Integration • Production Debugging & Observability • LangChain & LlamaIndex • Docker & CI/CD • React / Next.js • Structured Logging • End-to-End Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built and deployed **agentic LLM systems** for healthcare workflow automation; improved agent response stability ~25% and reduced hallucinations >30% under production constraints.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker, with structured logging and load simulation; reduced post-deployment defects ~30%.
- Designed **RAG systems** for clinical knowledge retrieval; improved contextual retrieval precision ~35%, reduced irrelevant retrieval >30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts and appointment histories; raised dataset reliability above 98% and cut model instability ~40%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP | Hyderabad, India

- Optimized SQL/T-SQL stored procedures powering the Order-to-Cash platform; reduced query latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%, release cycle time ~35-40%.
- Designed Grafana dashboards for performance monitoring; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time 60-70%.
- Deployed Flask-based API with web interface for non-technical staff; raised contextual accuracy >85%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated JSON contracts, RAG-grounded validation, and audit-ready reasoning traces. Owned end-to-end from design through production - same ownership model as Pallet's FDE deployment approach.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG

Dream Decoder - Multi-API Orchestration Platform

Full-stack FastAPI + React/TypeScript/Vite app integrating multiple AI providers with inconsistent APIs; built retry logic, response normalization, and intermediate prompt transformation layers. Improved contextual alignment ~**30%** and first-pass success rate ~**25-30%**.

FastAPI, React, TypeScript, Vite, Tailwind, multi-provider AI APIs

Manga Lens - Chrome Extension shipped, [Chrome Web Store](#)

Independently built and shipped TypeScript browser extension with 4 AI vision providers; multi-section capture pipeline, 7-day cache, 29-site support. High-agency delivery under sparse documentation.

TypeScript, Manifest V3, content scripts, multimodal AI APIs

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: TypeScript, Python, FastAPI, Flask, React, Next.js, SQL (PostgreSQL)

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, OpenAI/Anthropic/Google APIs

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers

Infra: Docker, Jenkins, Grafana, CI/CD, structured logging, observability, RESTful APIs

Domains: AI Agent Deployment, Enterprise Integration, Healthcare AI, Workflow Automation